

# Determining the Crystalline Structure of Metals through X-Ray Diffraction

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## Abstract

This experiment set out to measure X-ray diffraction patterns of various solid samples in order to describe their crystal structure at the atomic/molecular level. An already calibrated Siemens !!!!!MODEL NAME!!!! X-Ray diffractometer was used to send X-rays into Copper (Cu), Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples named A and B.

Any crystalline solid consists of a regular 3-dimensional packing of cells, the shape of which corresponds to one of fourteen Bravais lattices. These 3D patterns create multiple parallel planar layers within the crystal of various orientations. X-Rays travelling through these planes will diffract off the atoms and constructively interfere at the detector, creating a peak in photon count in the observed diffraction pattern. From that peak, Bragg's law  $n\lambda = 2d \sin \theta$  (where  $n = 1$  for a first-order plane) can be used to deduce the inter-planar distance  $d$  of the diffraction plane of the crystal. Using this inter-planar distance and the corresponding Miller indices, the Bravais lattice can be determined and the lattice parameter(s) calculated. The Cu and Ni pure samples and alloys were found to have a Face-Centered Cubic (FCC) Bravais lattice, with single lattice parameters [ ... ] respectively. The alloy samples of these two were found to have [ ... ]. The Sn and Pb samples were found to have a lattice of [...] and parameters [...] respectively, and their alloy [...], [...]. Finally, the mystery samples A and B were determined to be [...] and [...], with lattices [...] and parameters [...], respectively.

## Contents

|          |                                 |          |
|----------|---------------------------------|----------|
| <b>1</b> | <b>Introduction</b>             | <b>1</b> |
| <b>2</b> | <b>Methods</b>                  | <b>2</b> |
| 2.1      | Measurement Procedure . . . . . | 2        |
| 2.2      | Data Analysis . . . . .         | 3        |
| <b>3</b> | <b>Results</b>                  | <b>3</b> |
| <b>4</b> | <b>Discussion</b>               | <b>4</b> |

|            |                                                           |   |
|------------|-----------------------------------------------------------|---|
| Appendix A | Interplanar distance, Bravais Lattices and Miller indices | 6 |
| Appendix B | Supplementary Figures and Tables                          | 7 |

# 1 Introduction

The discovery of X-rays led to W.H. and W.L. Bragg researching the inner workings of X-ray diffraction. It was found that a solid can serve as a diffraction grating for X-rays of wavelengths similar lengths to the the inter-atomic distances. Similarly to how light diffracts, one can obtain a relation between the inter-planar distance  $d$ , the incident angle  $\theta$ , and the wavelength  $\lambda$ :

$$n\lambda = 2d \sin(\theta) \quad (1)$$

with  $n = 1$  for our purposes. Equation 1 is formally called Bragg's Law. Its utility becomes apparent when one ponders the crystalline structure of solids. Suppose the individual atoms of a crystal could be observed. While varying the angle the observer makes with the plane of the solid, there will be a moment where the atoms appear to line up. Given that a crystal is 3-dimension, the atoms will be aligned in parallel planes. Not all observed planes are equal however. Figure 2 shows a plane formed from a cubic arrangement of atoms. Notice that a plane could also have been formed from a face of the cube (in the y-z plane for example), or by aligning four diagonal atoms (the plane could be cutting through the y-axis and x-axis, for example). These planes are differentiated by their 'Miller index'.

In figure 2, [!!!EXPLAIN HOW TO GET MILLER INDEX!!!]

As briefly mentionned above, the atoms in Figure 2 are arranged as a cube. This set of arranged atoms is a called a cell, and they can borrow their symmetries from one of fourteen Bravais lattices, each grouped into one of seven crystal systems. Figure 1 shows these lattices and their groups. If an X-ray is sent to a solid with an angle  $\theta$  such that (1) is satisfied, then constructive interference occurs between the different rays, and a peak in the intensity may be observed at a detector placed  $2\theta$  with respect to the incident X-rays. From these peaks, the inter-planar distances can be obtained. This inter-planar distance is related to the Bravais lattice and Miller indices by various equations. For a cubic system, this relation takes the form of

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \quad (2)$$

where  $h, k, l$  are the Miller Indices and  $a$  is the length of the cube edge. Similar equations

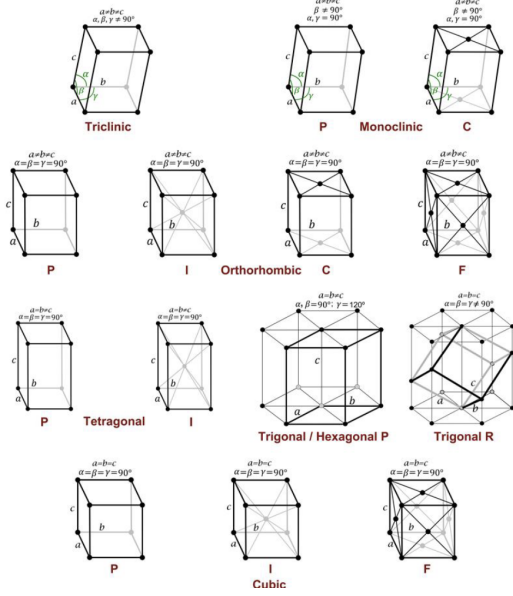


Figure 1: The fourteen Bravais lattices with their seven associated groups.

for different crystal systems are given in [Appendix A](#).

## 2 Methods

### 2.1 Measurement Procedure

A pre-calibrated Siemens diffractometer was used to produce X-Rays, send them at the samples, and detect the diffractions. The diffractometer consisted primarily of an X-ray tube, a sample mounting plate, and a photon detector. The X-ray tube used Copper  $K_\alpha$  radiation and was cooled using water. Two sets of filters of 0.6 mm were present on the X-ray tube and the detector, to prevent stray rays from leaving the tube and entering the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. The mounting plate kept the sample in place with a spring-loaded mechanism. The samples were centered by aligning the middle of the sample with a red line marked on the mounting plate. The diffractometer changes the  $\theta$  by rotating the mounting plate by some  $\Delta\theta$  and moving the detector by  $2\Delta\theta$ . A computer was connected to the diffractometer, with which

a .dql measuring parameter file was given to the diffractometer and data was obtained from

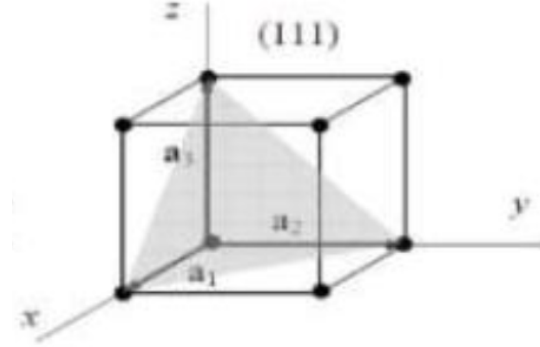


Figure 2: Plane corresponding to Miller index 111.

the dectector as .RAW files. The measuring parameters were an initial angle of  $2\theta_0 = 5.00^\circ$ , angle increments of  $\Delta\theta = 0.05^\circ$ , with 2.00 s at each step, with a final angle of  $2\theta_f = 110.00^\circ$ . The total measuring time is roughly 1 h 05m using the aforementioned parameters. Data acquisition was simple: Mount the desired sample onto the mounting plate, centering it with the red line. Then, on the computer, open the software program [!!!NAME!!!], select the desired parameter file, name the trial, and execute the job. After the job finishes, convert the .RAW file to .UXD. Finally, the .UXD files were transfered from the lab computer onto the experimentalists's for the data analysis.

## 2.2 Data Analysis

From the .UXD files of a sample, the plot of light intensity against  $\theta$  can be done. Using prominence 50, local maxima of light intensities can be located and indexed to find angles where diffraction in planes occur. Every peak in the plot is then fitted to a gaussian curve, centered at a local maxima, and every curve is summed to obtain a plot of summed gaussians of intensity against  $2\theta$ . From this, we extract mean values and variance.

## 3 Results

Three trials were done for Ni and Cu samples, though the data for the first Cu trial did not save correctly. Two trials were done for Pb, Sn, A and B. For each of the alloys (including different percentage of mixing) one trial was done. It was determined that the uncertainty due to random errors would be less that due to noise as is observable in the residuals of the gaussian sum fits by observing the Ni trials available in appendix [Appendix B](#) , figure [4](#). Therefore, only one trial was used for each sample. The fitted gaussian sums with residuals for the Cu and Ni samples and alloys are shown in the same appendix, in figure [5](#). From these, the inter-planar distances were calculated and are shown in table [??](#). These were plotted against expected results for all 3 types of cubic Bravais lattices, and the best fit was found to be the Face-Centered Cubic (FCC) lattice, as shown below in figure [??](#). The lattice parameters were then calculated and are shown in table [1](#).

Table 1: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard's law .

| Sample                            | $a_{\text{exp}}$ (Å) | $a_{\text{theory}}$ (Å) | $\Delta a$ (Å)       |
|-----------------------------------|----------------------|-------------------------|----------------------|
| Ni                                | $3.5223 \pm 0.0001$  | 3.5238                  | $-0.0015 \pm 0.0001$ |
| Cu                                | $3.6126 \pm 0.0001$  | 3.6149                  | $-0.0023 \pm 0.0001$ |
| Cu <sub>25</sub> Ni <sub>75</sub> | $3.5415 \pm 0.0002$  | 3.5467                  | $-0.0052 \pm 0.0002$ |
| Cu <sub>50</sub> Ni <sub>50</sub> | $3.5629 \pm 0.0003$  | 3.5696                  | $-0.0067 \pm 0.0003$ |
| Cu <sub>75</sub> Ni <sub>25</sub> | $3.5864 \pm 0.0002$  | 3.5924                  | $-0.0060 \pm 0.0002$ |

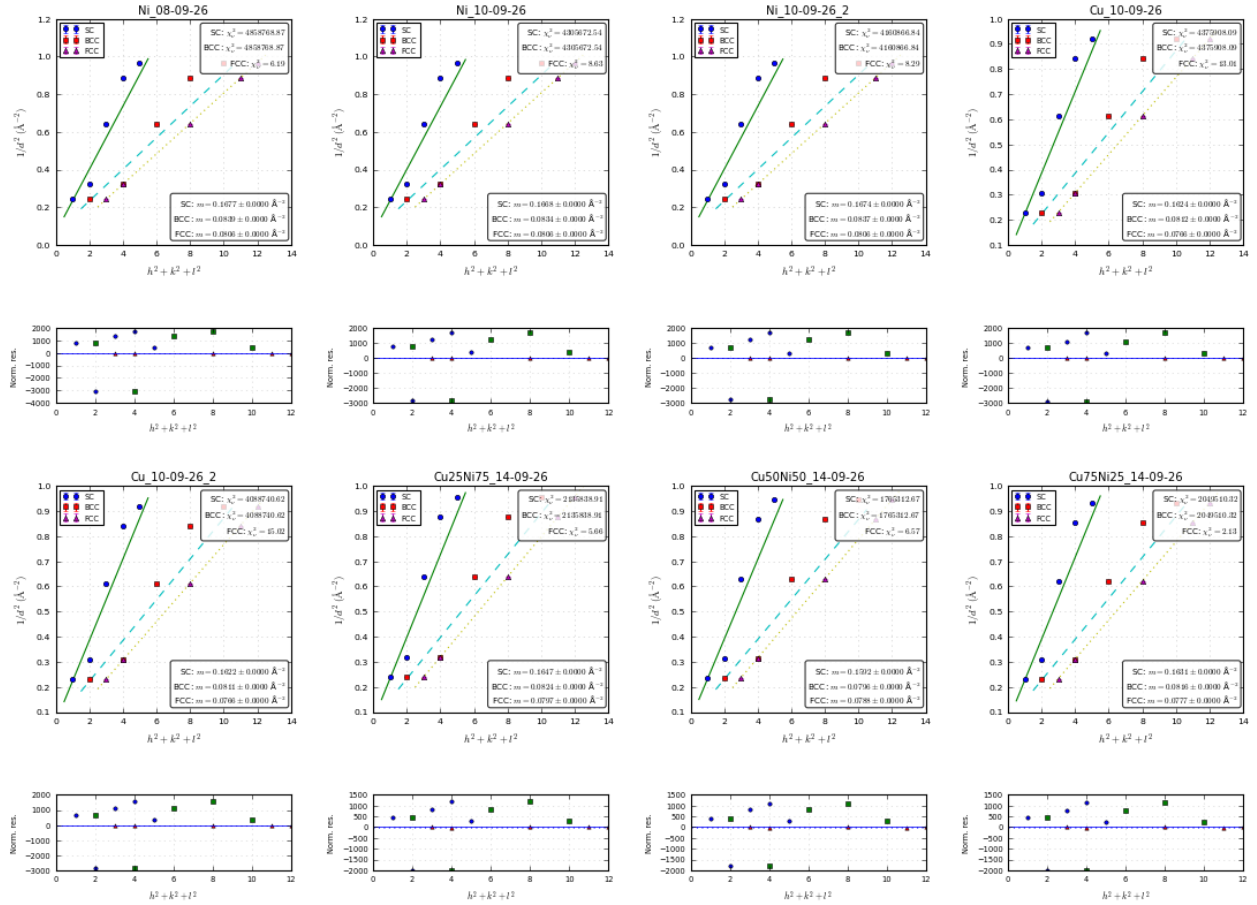


Figure 3: Determination of Cubic crystal structure and lattice parameter for Cu and Ni.

## 4 Discussion

# PLAN

## Accomplished

- Take data for all samples (pure Cu, Ni, Pb, Sn, CuNi alloys, PbSn alloys, and mystery samples A and B).
- Fit data using gaussian sum to find peak positions and uncertainties.
- Calculate inter-planar distances and uncertainties.
- Match inter-planar distances to Miller indices and cubic Bravais lattices for Cu, Ni, and CuNi alloys.
- Calculate lattice parameters and uncertainties for Cu, Ni, and CuNi alloys.

## To Do

- Produce gaussian sum fits for Pb, Sn, and PbSn alloys.
- Calculate inter-planar distances and uncertainties for Pb, Sn, and PbSn alloys.
- Determine how to match inter-planar distances to Miller indices for non-cubic Bravais lattices (Pb, Sn, and PbSn alloys).
- Determine Bravais lattices for Pb, Sn, and PbSn alloys.
- Calculate lattice parameters and uncertainties for Pb, Sn, and PbSn alloys.
- Buildin on the method developed above, determine the Bravais lattices and lattice parameters for mystery samples A and B.

## References

## Appendix A. Inter-planar distance, Bravais Lattices and Miller indices

For the following equations:  $d$  is the inter planar distance,  $a, b, c$  the inter-atomic distances,  $h, k, l$  the indices for a given Miller index (hkl).

Cubic:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:

$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:

$$\frac{1}{d^2} = \frac{4}{3} \left( \frac{h^2 + k^2 + hk}{a^2} \right) + \frac{l^2}{c^2}$$

Orthorhombic:

$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Vegard's law:

$$a_{\text{alloy}} = xa_{Cu} + (1 - x)a_{Ni}$$

## Appendix B. Supplementary Figures and Tables

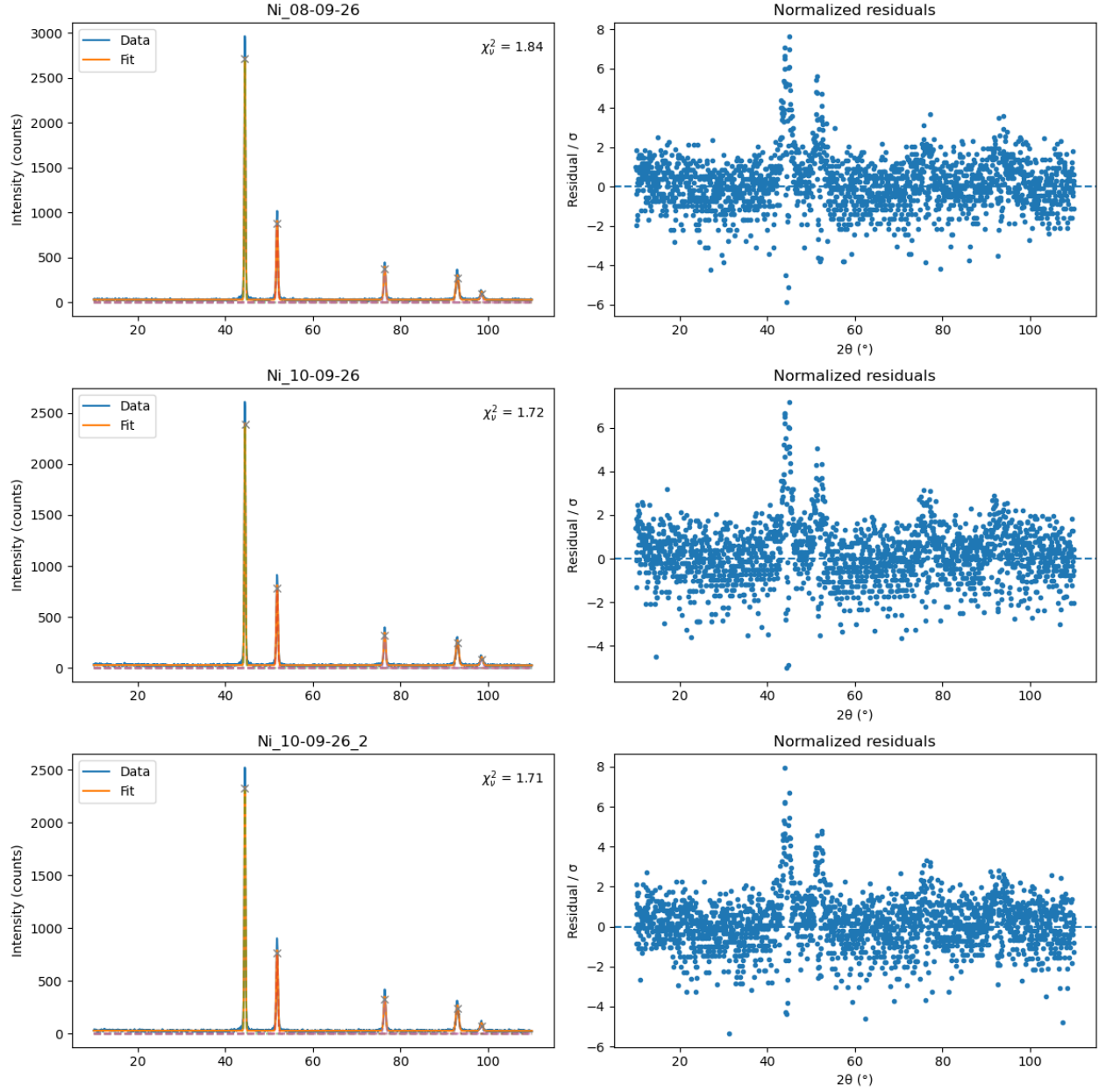


Figure 4: Gaussian sum fits of all pure Ni trials with residuals

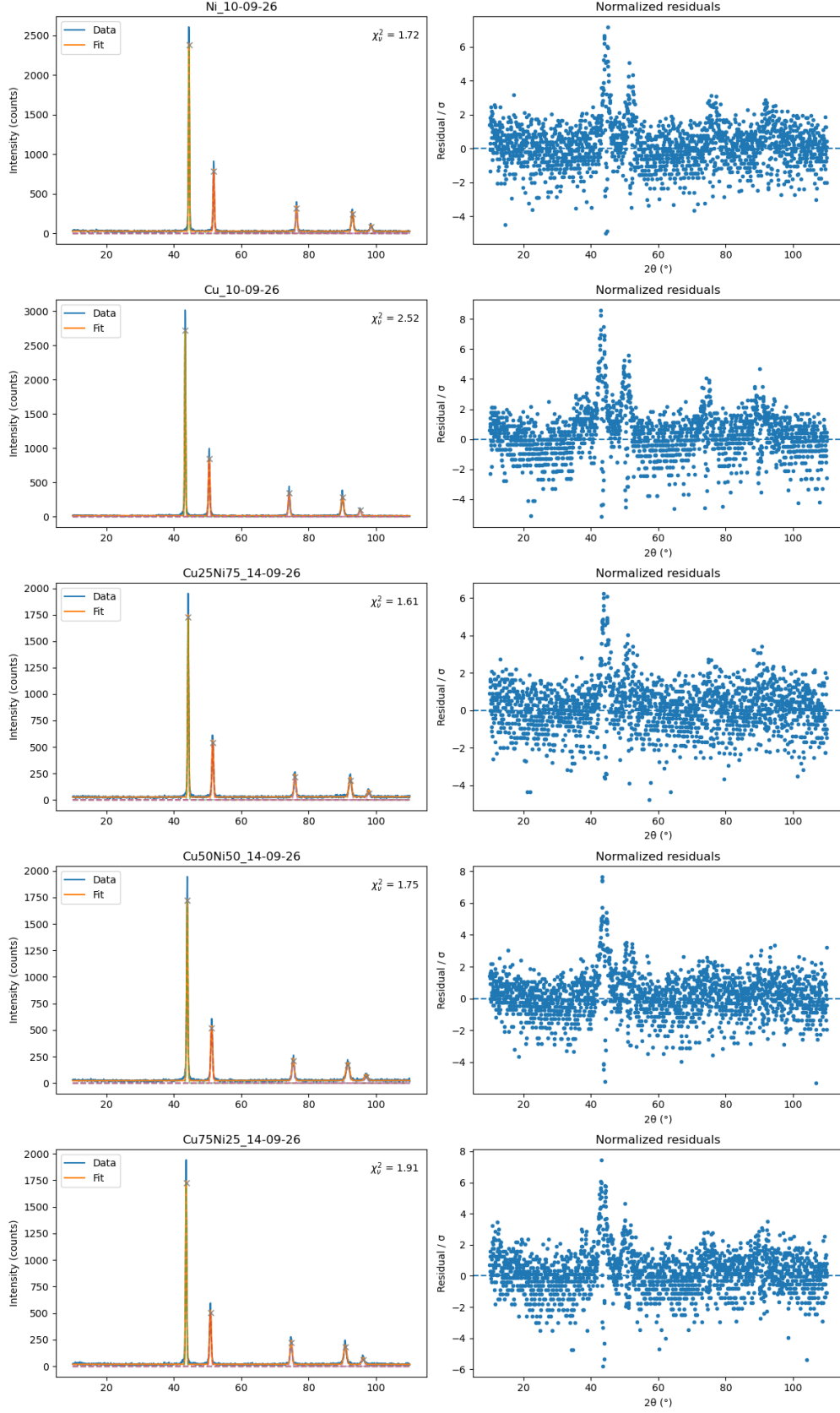


Figure 5: Gaussian sum fits with residuals of the Cu and Ni pure and alloy samples

| Ni trial 1                        |                      |                       | Ni trial 2                        |                      |                       |
|-----------------------------------|----------------------|-----------------------|-----------------------------------|----------------------|-----------------------|
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $44.4819 \pm 0.0011$ | $2.03512 \pm 0.00005$ | 1                                 | $44.4843 \pm 0.0011$ | $2.03502 \pm 0.00005$ |
| 2                                 | $51.8330 \pm 0.0023$ | $1.76246 \pm 0.00007$ | 2                                 | $51.8331 \pm 0.0024$ | $1.76245 \pm 0.00008$ |
| 3                                 | $76.3993 \pm 0.0041$ | $1.24563 \pm 0.00006$ | 3                                 | $76.3933 \pm 0.0045$ | $1.24571 \pm 0.00006$ |
| 4                                 | $92.9684 \pm 0.0059$ | $1.06221 \pm 0.00005$ | 4                                 | $92.9701 \pm 0.0062$ | $1.06220 \pm 0.00005$ |
| 5                                 | $98.4832 \pm 0.0109$ | $1.01694 \pm 0.00008$ | 5                                 | $98.5025 \pm 0.0127$ | $1.01679 \pm 0.00010$ |
| Ni trial 3                        |                      |                       | Cu trial 1                        |                      |                       |
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $44.4810 \pm 0.0012$ | $2.03517 \pm 0.00005$ | 1                                 | $43.3661 \pm 0.0011$ | $2.08487 \pm 0.00005$ |
| 2                                 | $51.8307 \pm 0.0024$ | $1.76253 \pm 0.00008$ | 2                                 | $50.4888 \pm 0.0024$ | $1.80618 \pm 0.00008$ |
| 3                                 | $76.4023 \pm 0.0045$ | $1.24558 \pm 0.00006$ | 3                                 | $74.2095 \pm 0.0043$ | $1.27687 \pm 0.00006$ |
| 4                                 | $92.9767 \pm 0.0062$ | $1.06214 \pm 0.00005$ | 4                                 | $90.0228 \pm 0.0054$ | $1.08915 \pm 0.00005$ |
| 5                                 | $98.4665 \pm 0.0135$ | $1.01707 \pm 0.00010$ | 5                                 | $95.2553 \pm 0.0097$ | $1.04266 \pm 0.00008$ |
| Cu trial 2                        |                      |                       | Cu <sub>25</sub> Ni <sub>75</sub> |                      |                       |
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $43.3667 \pm 0.0011$ | $2.08484 \pm 0.00005$ | 1                                 | $44.2433 \pm 0.0015$ | $2.04554 \pm 0.00006$ |
| 2                                 | $50.4882 \pm 0.0024$ | $1.80620 \pm 0.00008$ | 2                                 | $51.5406 \pm 0.0034$ | $1.77176 \pm 0.00011$ |
| 3                                 | $74.2000 \pm 0.0043$ | $1.27701 \pm 0.00006$ | 3                                 | $75.9226 \pm 0.0065$ | $1.25226 \pm 0.00009$ |
| 4                                 | $90.0104 \pm 0.0057$ | $1.08927 \pm 0.00005$ | 4                                 | $92.3239 \pm 0.0085$ | $1.06793 \pm 0.00008$ |
| 5                                 | $95.2715 \pm 0.0098$ | $1.04253 \pm 0.00008$ | 5                                 | $97.7855 \pm 0.0166$ | $1.02232 \pm 0.00013$ |
| Cu <sub>50</sub> Ni <sub>50</sub> |                      |                       | Cu <sub>75</sub> Ni <sub>25</sub> |                      |                       |
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $43.9763 \pm 0.0016$ | $2.05734 \pm 0.00007$ | 1                                 | $43.6499 \pm 0.0015$ | $2.07197 \pm 0.00007$ |
| 2                                 | $51.2219 \pm 0.0036$ | $1.78204 \pm 0.00012$ | 2                                 | $50.8455 \pm 0.0035$ | $1.79434 \pm 0.00011$ |
| 3                                 | $75.4112 \pm 0.0071$ | $1.25948 \pm 0.00010$ | 3                                 | $74.8037 \pm 0.0062$ | $1.26819 \pm 0.00009$ |
| 4                                 | $91.6044 \pm 0.0098$ | $1.07443 \pm 0.00009$ | 4                                 | $90.8289 \pm 0.0083$ | $1.08157 \pm 0.00008$ |
| 5                                 | $97.0016 \pm 0.0200$ | $1.02849 \pm 0.00016$ | 5                                 | $96.1327 \pm 0.0160$ | $1.03546 \pm 0.00013$ |

Table 2: Measured X-ray diffraction peak angular positions and corresponding interplanar spacings ( $d$ ) for the Ni, Cu, and Cu–Ni alloy samples.